GROUP RECOMMENDATION USING EXTERNAL FOLLOWEE FOR SOCIAL TV

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***Abstract*—Group recommendation plays a significant role in Social TV systems, where online friends form into temporary groups to enjoy watching video together and interact with each other. Online microblogging systems introduce the "following" relationship that reflects the common interests between users in a group and external representative followees outside the group. Traditional group recommendation only considers internal group members' preferences and their relationship. In our study, we measure the external followees' impact on group interest and establish group preference model based on external experts' guidance for group recommendation. In addition, we take advantage of the current watching video to improve context-aware recommendations. Experimental results show that our solution works much better in situations of high group dynamic and inactive group members than traditional approaches.**

Keywords- Social Media, Group Recommendation, External Expert, Context Filtering

I. INTRODUCTION

The rapid development of user generated content (UGC) and social network results in a large number of social media content. Media content related with social connections differs from traditional ones concerning that sources of social media tend to be generated from friends' sharing and recommender systems in social networks (SNS). User interaction in SNS continues to strengthen, followed by relationship expansion. Social TV brings a new trend of combining video streaming service with social network service, which has deeply influenced the way people produce and consume video content [1]. Some examples include the integration of Twitter updates during live video streaming [2] and Facebook applications that allow commenting while watching video content. Users tend to watch videos in groups instead of doing that alone [1]; as a result, the concept of group interactions in such systems has been boosted. Such group viewing provides great potential for users to find videos that interest members in the group, namely, group recommendation.

The composition of a group in such Social TV system varies from relatives, friends, classmates, and colleagues to even online concept users (e.g., public home page). And sizes of groups also vary from 3, 5 to 8 or even larger. Determining a group decision among alternatives with multiple attributes is a well-known problem that has been investigated for decades. Generally, there are two proposed solutions. One is to combine group members into a "pseudo-user", which can represent overall characteristics of the group [3]. The other is to merge recommendation results of individuals [4]. Existing work made use of different group decision strategies among which Average satisfaction, Least Misery and Most Pleasure are the most popular ones [5]. All methods are evaluated from the perspective of the content within group members.

Relationship within group members is also taken into account in group recommender system. Some measure user status within a group by evaluating relation strength with others. Superordinate users are even named "expert" and his characteristics can largely influence the preference of the whole group [5]. Others study the dissimilarity among group members [6], which also concern the content level. There are also some more studies on how each member contributes to reaching group consensus [9]. However, the existing approaches are facing the following problems. Firstly, group characteristics are affected not only by inside group members, but also outsiders, (e.g., idols). How to utilize them to predict interest of group, especially on social network with more openness and dynamic (e.g. Twitter)? Secondly, how to satisfy the whole group when encountering sparse relation among group members?

To address the above problems, in this work, we consider influence of external experts; aiming at performing group recommendation under situation of more openness relation with people outside the group, lower degree of close relationship within group and much higher dynamic group formation.

Our mainly contributions answer the following questions in group recommendation. (1) Can group recommendation benefit from external experts? (2) In what kind of groups do external experts perform the best? (3) How useful is context filtering in group recommendation? (4) Will groups with various sizes have different recommendation satisfactory?

We name the user out of a group who can contribute to the group recommendation as "external expert". Firstly, we determine the scope of external expert by measuring their number of fans to evaluate their vitality. We also take into consideration of external experts' relationship with inner group members by making use of fans and idols' relation. Before external experts making real contribution to describing group preferences, we cluster and measure various significance of external experts. In addition, context of the current viewing video is utilized to finalize the recommendation results.

The rest of the paper is organized as follows. We model group preference using external experts in Section 2. Section 3 presents our cascading group recommendation algorithm. In Section 4, we conduct comprehensive experiments to evaluate the effectiveness and performance of our design. Finally, we conclude the paper in Section 5.

II. GROUP PREFERENCE MODELING BASED ON EXTERNAL EXPERTS

A. *Describing Group Interests Using External Experts*

Since introducing external experts into groups brings plenty of uncertainties, we need to analyze relationship between external experts and inner group members to see whether they can describe the interests of members in the group. We first determine which external experts are selected, and then we measure their description ability.

Determine Scope of External Experts: When considering external relations, we measure the two-way relationship which means on one side is a group member, on the other side is an external user and both are followees to each other. With source data from API of Tecent Microblogging Service, an illustration is shown in Table 1, we observe that the two-way relation is basically between real life social circles such as friends, classmates, colleagues and so on. Though this kind of external users have relatively close relation with group members, they are not qualified enough to be an "external expert" for most of time, i.e., they are neither active enough nor sharing highly similar interests with the group members. Instead, their inputs to the social network are mainly greeting each other, getting to know each other's current situation and giving some responses. So users who have two-way relationship with group members can be removed when considering an outsider's impact on interest of group.

TABLE I. COMPARISON IN AVERAGE FANS' NUMBER FOR EXTERNAL USERS HAVING ONE-WAY RELATION AND TWO-WAY RELATION WITH GROUP MEMBERS

Group Size	No. of Avg. fan in Two-way Relation (TR)	No. of Avg. fan in One-way Relation(OR)	Ratio= OR/(TR +OR)
3	135	1561685	99.991%
5	148	1073036	99.986%
5	14282	1849390	99.992%
8	112	1574386	99.993%
8	267	1152384	99.977%

Common Behavior Measurement on External Experts: We take into consideration of guiding significance from external users. After removing those who have two-way relationship with group members, we move to measure how the remaining external users influence group interests. We classify them into two categories. One includes who we called "Celebrity", for they have more than 1million fans on the social network and are generally called "Stars" in varies fields. Most of them rank among the top 5000 users by statistics from Tencent microblogging system according to the number of fans. And for the other, we refer to those who are relatively not so famous, and they have hundreds or thousands of fans, but they are still popular in a certain region with certain reason both in real life and on social network, such as a free occupational psychologist or a constellation researcher.

Originally, we think that influence of celebrities' behavior has much greater effect on those of group members than the "General famous" users. Then we measure 5 groups with varies scale of 3, 5 and 8 people, and also meets different social relation scenarios. We define **Common Behavior Rate** (**CBR**) to measure relation between a group member and an external expert. We first count microblogs which have been primary, forwarded or commented on by both of them. Then the number is divided by the total number of microblogs related with the group member to get the CBR which means CBR is specific to group members. The "common behavior rates" are presented in Table 2.

TABLE II. TOTAL CBR FOR ONE-WAY EXTERNAL EXPERTS

Group No.	1	2	3	4	5
CBR (%)	50.67	37.60	42.37	56.86	53.16

We observe that "Top Star Phenomenon" do not really exist, for external users belong to different fans' quantity interval do not show much difference from each other. It means we can not measure an external user's influence on group's interest according to his fans' number. We further analyze the statistics of a group whose members are all fans of Chinese popular Sports Star Liu Xiang. Result is shown in Fig. 1.

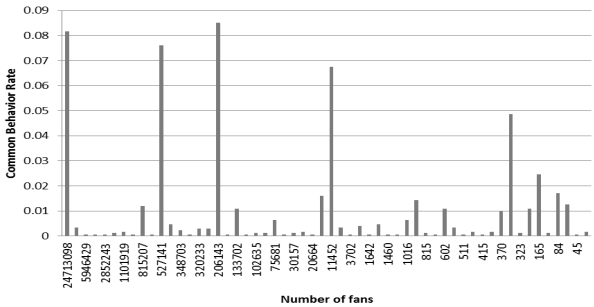

Figure 1. Relation Between Fans' Number and CBR

After checking external users of the group who rank among the top rate, we find they are Liu Xiang himself, Shi Dongpeng who are Liu Xiang's teammate, Liu Xiang's coach and Tencent Sports reporter. And other external users with high quantity of fans are Xiao Cui Talking Show and so on. It can be drawn that a high Common Behavior Rate on microblogging such as forwarding the same microblogging or making comments on it can significantly indicate they share same interests while number of fans simply has little contact with interest sharing.

TABLE III. EXTERNAL PROFILE AND INTERNAL PROFILE SIMILARITY

Group No.	1	2	3	4	5	Avg.
ProfileSim	0.90	0.93	0.84	0.94	0.77	0.88

External and Internal Profile Similarity: We define **ProfileSim** as similarity between whole External Experts' profile and whole Group Members' profile by calculating their history behaviors on microblogging in 10 dimension vectors on behalf of topic model. (Detail algorithm will be explained in "Group Preference Model".). Table 3 shows that ProfileSim is quite high and we can use External Profile to represent Internal Profile for groups whose members are inactive and cannot be described by their own history behavior.

B. Group Preference Model

Facing strong group dynamic, weak closeness among group members and relatively greater openness in group recommendation, we come up with a solution, with key point of introducing external relationship coming from group members' idols to a group to describe the preference of the group.

DEFINITION (EXTERNAL EXPERT PROFILE): Profile to a group G from the aspect of external experts is denoted as $preEx(G,i)$. We classify external experts into clusters and each cluster's preference is described as exP_i , together with its weight w_{ei} which represents the expert cluster's importance among whole external experts. We carry out the external experts profile as follows:

$$preEx(G,i) = \frac{\sum_{i=1}^{N} exP_i * w_{ei}}{\sum_{i=1}^{N} w_{ei}} \tag{1}$$

1) External Experts Preferences: We first determine range of experts. Based on measurement results, we remove external users with two-way relationship. For the others, we call them "External Experts". Experts' behavior on microblogging is more close to group members' interest while friends care more about each other's situation.

As group members are very interested in social media content generated by their idols, using external experts' history behavior can well reflect group members' interests. We study how to describe External Experts' Characteristics. As each expert's history is composed of microblogging he/she posts and shares, we move to abstract features of each microblogging with videos. In this work, we use topic model clustering algorithm [7] to find out a video's eigenvectors. We crawl titles and tags of 45,470 videos from top 10 famous video sharing sites in China. After words segmentation by LDA and data training, we obtain 10 topics by Topic Model cluster which have probability statistical significance and each video has a 10-dimension vector on them. Results show words contained in each topic cluster can be seen as a class of words. Words between classes can also be recognized as differentiated from each other to some degrees. It demonstrates that using 10 basic topics can generally describe the characteristics of a video with data sets on this scale.

2) External Expert Weight of Significance: With such sparse relation within a group, each member may receive different information or contact with people from different areas or fields. So in few cases each external expert can be followed by more than one group members which are often the case with small groups. But different experts can be categorized into same category, such as sports stars or current affairs commentators. By clustering external experts, we can easily find group members' common interests. We use K-means clustering algorithm [8] to classify external experts into K=15 clusters, which is an empirical value coming from the number of categories of celebrities on Tencent Official Microblog.

Each expert cluster needs a weight to describe their different contribution to internal group members' interest among the whole expert group. Here we set two kinds of weights. One is **Weight of Followed Number** since we know that idol-fans relation can get people with similar interest together. The other is **Weight of Common Behavior** as we have measured that common behavior is an important indicator of similar interest. So we count total number of idol-fans relation and common behavior between an expert cluster and inner group members. Then summate them to be **Weight of Cluster** to describe the significance of the expert cluster among whole expert group.

III. EXTERNAL EXPERT BASED GROUP RECOMMENDATION

A. Proposed Framework

Our proposal is composed of two Steps: The First Step Recommendation is conducted by External Experts' contribution to group preference. First we find all the one-way follow relation out of a group and regard them as external experts. By clustering external experts into clusters, we get an orthocenter for each cluster to represent the cluster.

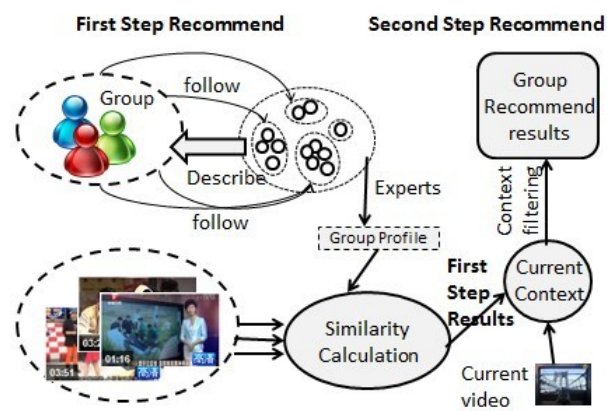

Figure 2. Framework of Our Group Recommendation Algorithm

We also attach each cluster with specific normalized weight. After weighted average, we achieve the feature of a unique "virtual external expert" which behaves as overall

external contributions to group recommended. Then we calculate its similarity with candidate video list.

In the second step, we use context filtering to further improve the relevance of recommendation.

B. Initial Recommendation with Group Preference

With Group Preference Model, we get a "virtual external expert" profile for a group. Then we calculate each item's similarity with it in candidate list. The similarity of an item to a group G from the aspect of external experts, celebrities in microblogging followed by group members for example, denoted as $simEx(G,i)$. As external experts profile describes as $preEx(G,i)$, and v_i represents a video in candidate list. We carry out the similarity computing formulation as follows:

$$simEx(G,i) = v_i * preEx(G,i) \tag{2}$$

After sorting the results, we get the top 100 items as the outcome of first step group recommendation.

C. Current Context Filtering

Existed algorithms are generally conducted by analyzing historical data from users which seldom concerns about influence of current context, especially in group recommender systems. Actually, current context do have impact on users' choice in subconscious. We make use of current context to filter preliminary group recommendation result. Feature of current context is described as p_c , each recommending candidate is denoted as v_i . The similarity of current context to each item is denoted as $sim(p_c, v_i)$. We define filtering index for each item is fl_i. So we have context filtering formulation as follows:

$$fl_i = sim(p_{c,} v_i) \tag{3}$$

After sorting the results, we get the final group recommender results.

With the above two steps, we can get a comprehensive group recommend proposal which can fully meet needs of a more open social network with features of great dynamic and sparse tightness.

IV. PERFORMANCE EVALUATION

In this section, we conduct experiments based on real-world groups to evaluate our group recommendation solution. We compare our design with three state-of-the-art group recommendation algorithms: Average Satisfaction, Least Misery and Most Pleasure.

A. Experiment Setup

Experimental simulation scenario: Suppose the participant is watching video together with a group of people on microblogging social network. When the playing video comes to an end, members in the group are going to decide which one to watch together in the next. Then the participant should give his/her own opinion on the recommended video presented, by marking scores ranging from 1 to 5. The recommended videos are generated by our design and the other three group recommendation strategies.

We invited 16 groups (70 individuals, with an average age of 25) to form various scale of groups including 8 groups of 3, 6 groups of 5 and 2 groups of 8. These participants are college students and graduates with various profession backgrounds. In each group, users seldom know each other, especially about each other's interest. So we invited them to perform the test and made fair comparison with other methods. Then we recognize their internal and external social relations by using Tencent microblogging's APIs. We get each group member's idol list, fan list, history behaviors, including primary, forward or comments, etc. By analyzing the data, we can recognize external experts of each group and their relationship with internal group members through the procedure of collecting social relations from each participant.

We develop a website to be our experiment platform and selected the hottest 3,500 videos in 7 categories for the whole month of November 2011 from Youku—the most popular Video Website in China—as candidate lists for group recommendation. The experiment is divided into two parts. In **Part One**, on the purpose of evaluating whether context filtering can improve relevance of recommendations, we set a contrast experiment by recommending videos without using context filtering in our solution. In **Part Two**, we suppose the current context influences people's decisions about what they want to watch next, so we introduce context filtering. The selected test videos are divided into two categories. One is **Similar Videos** as they meet the demand that each group member have similarity with the video ranging from 0.7 to 1. The other is **Dissimilar Video** as each group member has similarity with the video ranging from 0 to 0.3. Then participants are going to do six tests. In each test, we present a selected video (the selection method will be illustrated later) to let participants watch it which imitates the group are watching the video together in an online social room. Then they are asked to mark scores for 8 recommended videos which comes from four methods with top two videos for each on their recommend lists. All recommended videos are filtered by current video context.

B. Evaluation Criteria

In the experiment, for each group, we have 7 tests. Except for the first one, the remaining 6 need participants to watch a video and then mark for the recommended 8 videos. For each recommended video, each group member should mark a score from 1 to 5 and a same score cannot be marked for more than three times. And we can get an average score for each video in one group. We define the **HIT RATE (HR)** as measures of merits of different target recommend algorithms. So HIT RATE is defined as follows: After we sort average score for 8 videos in one test, we pick out the top 3 ones and calculate ratios for each of four strategies' hit number out of total recommend number. Specifically, as Strategy "A" recommends 2 videos in one test, if the top 3 videos according to their average score contains 1 from strategy "A", then A's HR=50%; if 2 videos are from it, then

its HR=100%, of course, if the top 3 contains none from "A", then its HR=0%. By comparing HR, we can well evaluate different group recommender strategies.

C. Experiment Results Analysis

1) Context Filtering Evaluation: To evaluate whether the current context filtering really improve the effectiveness of our proposal, we design the first test in each group experiment. As we have mentioned, each recommended video will be marked by all the current group members so we take average and RMSE of each video's score in each group as assessment criteria. We divide all experimental groups into two categories. One is using context filtering while the other is not. The results can be seen in Fig. 3.

We can see that after adopting current context filtering, average score increases which indicates overall group satisfaction increases. The variance becoming lower which means that uniformity of group interest improved through context filtering. It proves that using current context filtering to help group recommendation is valid.

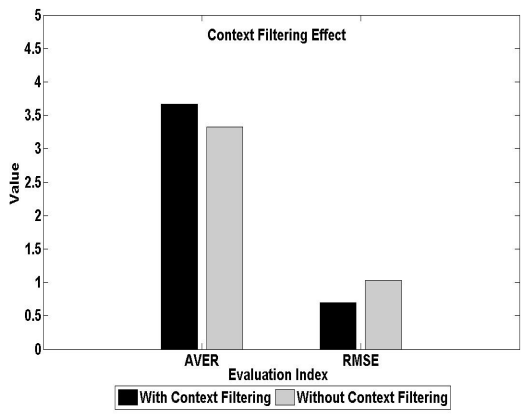

Figure 3. Context Filtering Effect

2) Group Scale Evaluation: We next investigate impact of different group sizes on group recommendation. In our experiments, we have groups in three different sizes: 3, 5 and 8. We use **HIT RATE** to evaluate performance of four methods in various group sizes. The results shown in Fig. 4.indicate in cases of different group sizes, our method can always perform much better than the other three which also bear some fluctuations. Detailed analysis is shown in Table 4.

TABLE IV. AVERAGE MICROBLOGGING AND FOLLOW NUMBER

Group Size	3	5	8
Avg. microblogging number	32.42	21.51	86.88
Avg. follow number	90.00	54.24	50.83

We can see with average follow number increasing, advantages of our method become more and more obvious which further verify the beneficial effect of external expert guidance. Specifically, compared with groups of size 8, average microblogging number of groups with 5 members is much lower, but its average follow number is a bit higher which brings better overall satisfactory on group recommendation results. This well explains why our solution performs much better than the other three which only make use of inner group features to recommend items.

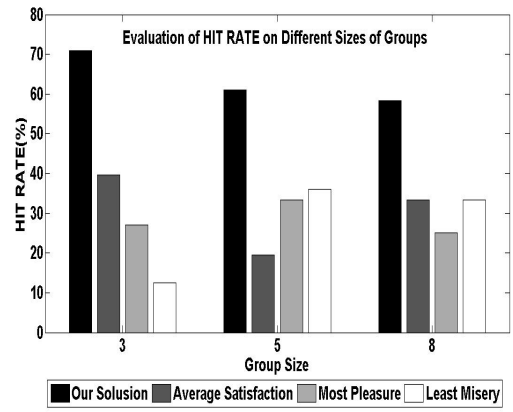

Figure 4. HIT RATES under Different Group Sizes

3) Group Vitality Evaluation: We evaluate the impact of external experts on group recommendation, which is our key point. As we have illustrated, the significance of concerning external experts is to contribute their interests to guide group interest when group members are inactive or new registered users. So we sort participant groups by their average microblogging number. And we divide them into three intervals with each interval length for 50. Then we evaluate **HIT RATE** of four strategies in each of the intervals, including (0,50), (50,100), (100,+∞). The performance is illustrated in Fig. 5. As the Fig. 5 shows, in all the three intervals, our solution performs excellent and far better than the other three group recommendation strategies. In more detail, in interval of (0, 50), group members are not active. In this case, superiority of our solution's is obvious thus it can be a good strategy to deal with cold start issue which is always annoying. In interval of (50,100), with the increasing number of users' microblogging, our solution still significantly performs better than other strategies. As the other three strategies' effectivenesses rise a little, HIT RATE for our solution draws down relatively which is quite reasonable.

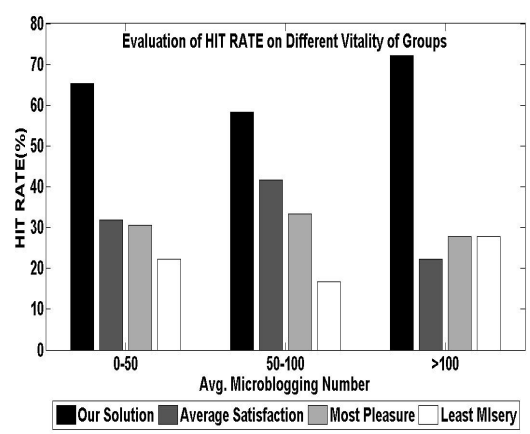

Figure 5. HIT RATES under Different Group Vitalitie

When comes to the interval of (100, +∞), users have gradually become active users. As we can see in Table 5, average external experts' number in this stage is much higher than the former two stages.

TABLE V. AVERAGE EXPERTS' NUMBER

Avg. microblogging number	0-50	50-100	>100
Avg. No. of Experts	53.22	69.20	141.08

So we can get that our solution concerning external experts is much more effective than the other three strategies on the whole and can well solve the cold start problems while others cannot. Additionally, with one user become more and more active or use the social network for longer time, our solution can be better and better than the others.

4) Impact on Test Video Selection: We can see from Fig. 6, when test videos are similar to group members' preferences, context filtering impact is counterbalanced after cascading. While for those dissimilar ones, better HIT RATE indicates the filtering really takes effect. This brings benefits on recommendation when looking for diversiform videos.

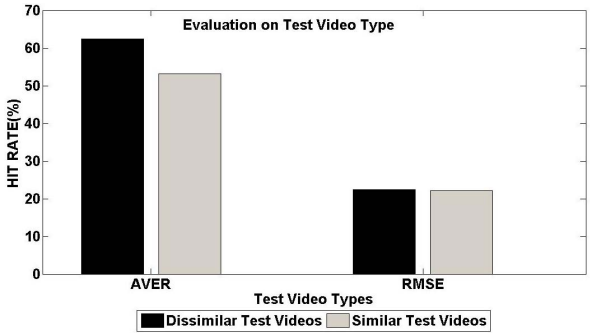

Figure 6. HIT RATES under Different Test Videos

V. CONCLUSION

In this paper, we make use of external experts to describe the interests of a group. Using the external expert based group preference model, we recommend videos to members in the group. In our design, the video which users are watching is referred to as context information. Experiment results show that our design achieves significantly better performance in situations of high group dynamics and inactive group members than traditional methods under different group sizes. Our scheme can also be applied to other situation, such as group recommendation on e-commerce if we use microblogs with e-commerce content instead of video content. For future work, we should combine external and internal factors to see how a comprehensive method enhances group recommendation. We also need to research on larger groups as group sizes in our design are relatively small.

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